Project Proposal

ECOSYNC

"Enhancing Comfort, Conserving Power, Innovating the Future"

Intelligent AC Temperature Regulator

SocketBurners

Aashir MRM
220004M

Muftee MMM
220399B

Munavvar MAA
220405T

Nadha MI
220409J

EN1190 – Engineering Design Project

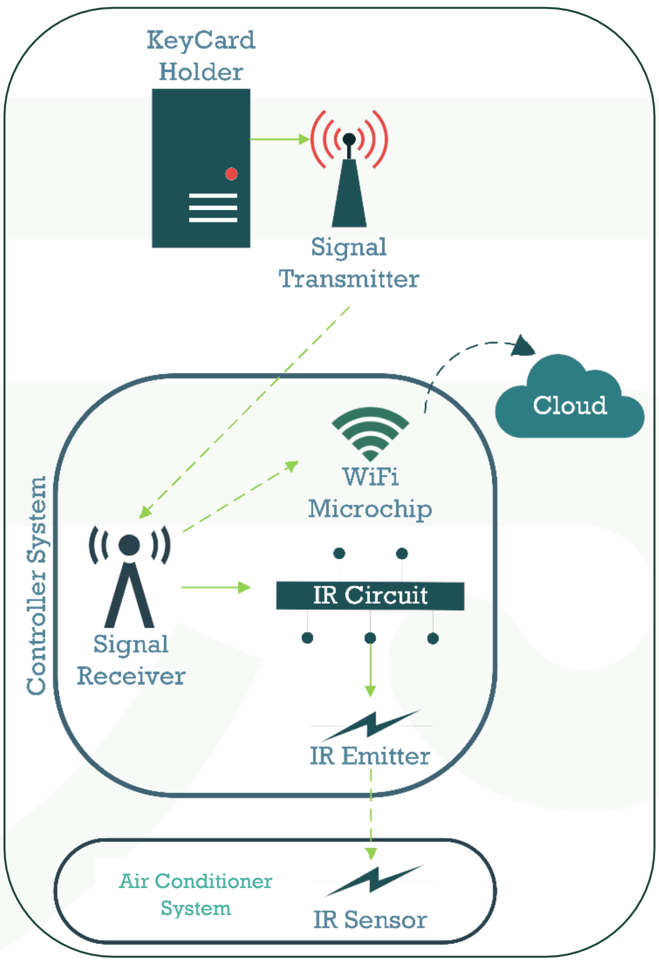
Introduction

One of the major concerns in today's industrial world is the energy deficit. This brings various challenges to energy consumption due to rising energy costs & wastage.

Hotels have implemented a key card system that automatically switches off the electricity when guests leave their rooms to conserve energy. However, upon returning to their rooms, guests often experience discomfort due to the temperature rise and must manually adjust the air conditioner. However, some hotels keep the air conditioner running even after removing the key card to ensure guest comfort, leading to significant energy wastage.

As an effective solution for this problem, we came up with the idea: "ECOSYNC."

which adjusts the AC level without turning it off based on the presence of guests as a practical solution to optimize energy usage while maintaining guests' comfort.



Concept

The standard temperature range for air conditioners when the user is present in the room is typically between 18 to 20 degrees Celsius.

When the keycard is removed, the ECOSYNC system will automatically adjust the AC temperature to a higher level around 27-28 degrees Celsius. This feature can help save energy consumption by up to 50% - 70%. Once the Keycard is inserted again, the system will automatically return to the initial lower temperature levels.

It prevents user discomfort and at the same time conserves energy.

Implementation Objectives

1. Check whether the keycard is inserted or not.
2. Send the signal to the controller's receiver.
3. Emit the IR rays from the controller to the AC's IR receiver to change the temperature.
4. Send data to the Cloud Network.

Resources

1. PCB with Controller
2. 2 Transceivers
3. An IR Emitter
4. A Wi-Fi Microchip
5. Power Supply

Expected Scope and Outcome

Target Scope:
We mainly focus to establish a market in the luxury hotel field where the keycard system is in use.

Expected Outcome:
Reducing nation's energy wastage.
(Offering high quality customer service in the hotel industry)

ECOSYNC

Product Description

Ecosync is a smart A/C regulator designed to be used in luxurious hotels. Its primary function is to maintain the A/C temperature at an optimum level (i.e: A level at which both user comfort and energy usage are considered)

Problem Validation

- ***Sri Lanka's point of view.***

Sri Lanka is going through unprecedented times. The economic crisis continues to worsen on one side and on the other side we have a shortage of energy supply to meet current energy demands. Amidst this status quo, it's imperative for us as a nation to do everything possible to preserve energy and to impose regulations to mitigate energy wastage. Therefore, it is in the best interest of the country to have regulations in place to control energy usage in luxurious hotels. In this context, our solution comes in handy since our proposed solution cuts down energy used by the air conditioning system by 50 %. This creates a huge incentive for hotels to consider our product.

- ***Energy Regulations set by the government.***

Hotels must follow energy regulations that aim to promote sustainability and reduce environmental impact. These regulations cover different aspects like energy efficiency standards, renewable energy utilization, and energy conservation measures. Hotels are required to meet specific standards for lighting, AC systems, and appliances.

- ***Global point of view.***

Sustainable Development Goal 7 (SDG7) calls for "affordable, reliable, sustainable and modern energy for all" by 2030. The world collectively is promoting measures to achieve sustainable energy. Through our product, we can conserve valuable energy.

In addition, hotels are also incentivized by this. They can earn tax benefits, be nominated for awards, etc… for demonstrating sustainable practices. This is a win-win for our proposed solution.

- ***Luxury Hotels' point of view.***

The hotels that are under consideration here are in the business for money. They look for ways to increase their profits. While having to abide by government regulations, promote an eco-friendly image, and strive to achieve UN sustainability goals are within the Hotel's mandate, there is something else that interests them more, which is profit. This is where we come into the picture.

As per the tariff approved by the Public Utilities Commission of Sri Lanka, hotels are charged 25 LKR/kWh for energy consumption between 0-180 kWh, and 36 LKR/kWh for energy consumption above 180 kWh. This leads to a fourfold increase in the monthly charge.
Reference: https://www.pucsl.gov.lk/electricity/tariff/industrial-consumers/

By cutting down the energy consumption by 52% using our product, we can bring down the costs by more than 50%. This is extremely beneficial for these hotels, which in turn creates demand for our product.